Labsheet 8 SECS05-AS08

Mastery in Data Analytics and Visualizations and Career Advancement

Khushi Choudhary CU240251564

In [8]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import classification_report, confusion_matrix, accuracy_sc

df = pd.read_csv("Telco-Customer-Churn.csv")
print("Dataset Shape:", df.shape)
print("\nFirst 5 rows:")
print(df.head())
```

```
Dataset Shape: (7043, 21)

First 5 rows:
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No

  TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling  \
0          No          No              No  Month-to-month              Yes
1          No          No              No        One year               No
2          No          No              No  Month-to-month              Yes
3         Yes          No              No        One year               No
4          No          No              No  Month-to-month              Yes

                PaymentMethod MonthlyCharges  TotalCharges Churn
0            Electronic check          29.85         29.85    No
1                Mailed check          56.95        1889.5    No
2                Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4            Electronic check          70.70        151.65   Yes

[5 rows x 21 columns]
```

In [9]:
```
# Data Preprocessing
# Drop non-predictive identifier column
df = df.drop(columns=['customerID'])
# Convert 'TotalCharges' from string to numeric, coercing spaces/errors to NaN
```

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
# Fill missing TotalCharges values with the median column value
df['TotalCharges'].fillna(df['TotalCharges'].median(), inplace=True)
# Encode binary target variable 'Churn' (Yes -> 1, No -> 0)
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})

# One-Hot Encode remaining categorical features
categorical_cols = df.select_dtypes(include=['object']).columns
df = pd.get_dummies(df, columns=categorical_cols, drop_first=True)

# Separate features (X) and target variable (y)
X = df.drop(columns=['Churn'])
y = df['Churn']

# Split Dataset into Train and Test Sets (80:20)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
```

```
/tmp/ipykernel_14928/4235312855.py:7: ChainedAssignmentError: A value is being se
t on a copy of a DataFrame or Series through chained assignment using an inplace
method.
Such inplace method never works to update the original DataFrame or Series, becau
se the intermediate object on which we are setting values always behaves as a cop
y (due to Copy-on-Write).

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.meth
od({col: value}, inplace=True)' instead, to perform the operation inplace on the
original object, or try to avoid an inplace operation using 'df[col] = df[col].me
thod(value)'.

See the documentation for a more detailed explanation: https://pandas.pydata.org/
pandas-docs/stable/user_guide/copy_on_write.html
  df['TotalCharges'].fillna(df['TotalCharges'].median(), inplace=True)
/tmp/ipykernel_14928/4235312855.py:12: Pandas4Warning: For backward compatibilit
y, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. T
his behavior is deprecated and will be removed in a future version. Explicitly pa
ss 'str' to `include` to select them, or to `exclude` to remove them and silence
this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-mig
ration-select-dtypes for details on how to write code that works with pandas 2 an
d 3.
  categorical_cols = df.select_dtypes(include=['object']).columns
```

In [11]:

```
# Train Decision Tree Classifier
model = DecisionTreeClassifier(criterion='gini', max_depth=4, random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

# Model Performance Evaluation
print("\n--- Model Evaluation Metrics ---")
print(f"Accuracy:  {accuracy_score(y_test, y_pred):.4f}")
print(f"Precision: {precision_score(y_test, y_pred):.4f}")
print(f"Recall:    {recall_score(y_test, y_pred):.4f}")
print(f"F1-Score:  {f1_score(y_test, y_pred):.4f}\n")
print("Classification Report:")
print(classification_report(y_test, y_pred))

# Plot Confusion Matrix
```

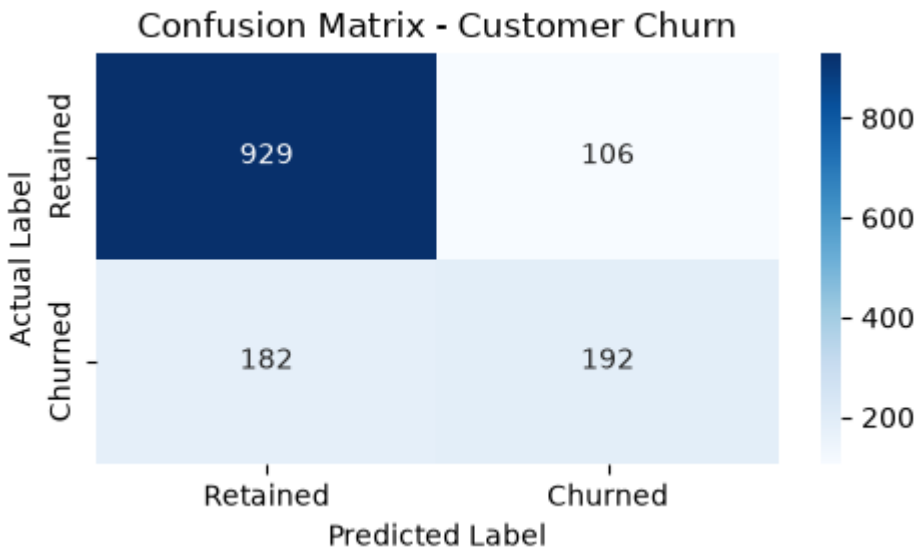
```
cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(5, 3))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Retained', 'Chu
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')
plt.title('Confusion Matrix - Customer Churn')
plt.tight_layout()
plt.show()
```

```
--- Model Evaluation Metrics ---
Accuracy:  0.7956
Precision: 0.6443
Recall:    0.5134
F1-Score:  0.5714

Classification Report:
              precision    recall  f1-score   support

           0       0.84      0.90      0.87      1035
           1       0.64      0.51      0.57       374

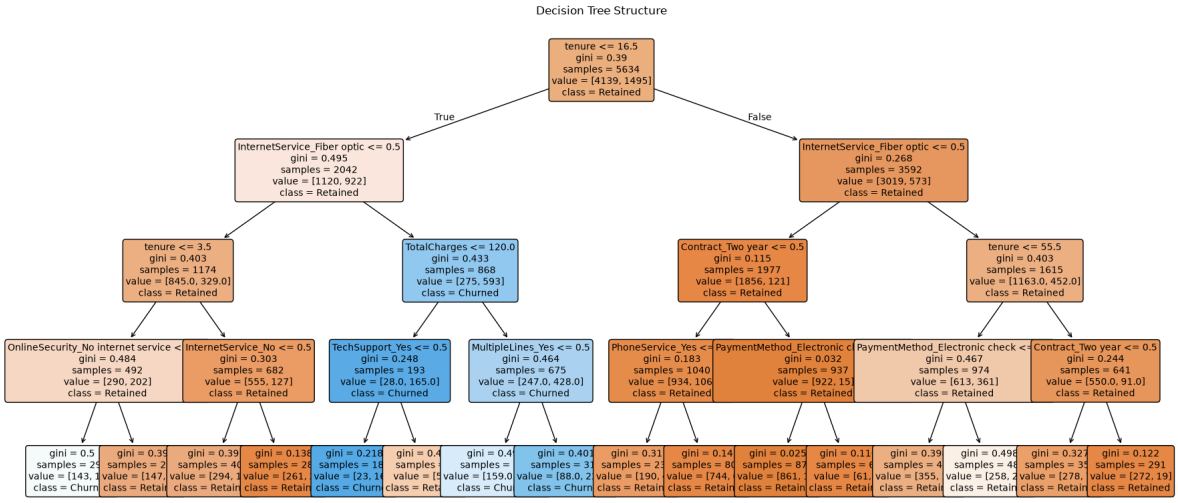
    accuracy                           0.80      1409
   macro avg       0.74      0.71      0.72      1409
weighted avg       0.79      0.80      0.79      1409
```



In [14]:
```
# Visualize Decision Tree & Feature Importances
plt.figure(figsize=(18, 8))
plot_tree(
    model,
    feature_names=X.columns,
    class_names=['Retained', 'Churned'],
    filled=True,
    rounded=True,
    fontsize=10
)
plt.title("Decision Tree Structure")
plt.tight_layout()
plt.show()

# Extract and plot top feature importances
importances = pd.DataFrame({
    'Feature': X.columns,
```

```
    'Importance': model.feature_importances_
}).sort_values(by='Importance', ascending=False)
plt.figure(figsize=(8, 4))
sns.barplot(x='Importance', y='Feature', data=importances.head(10), palette='vir
plt.title('Top 10 Most Influential Features in Predicting Churn')
plt.tight_layout()
plt.show()
```




```
/tmp/ipykernel_14928/995030745.py:21: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.barplot(x='Importance', y='Feature', data=importances.head(10), palette='vi
ridis')
```

